

Follow-up Assignment:

From Binary Files to H2: Persisting Encoded Circles in Java

December 2, 2025

1. Project preparation

- Create a new project with Gradle as the build system.
- Open the file *build.gradle.kts* and in the *dependencies* section add the entry `implementation("com.h2database:h2:2.2.224")`. Synchronize the Gradle changes.
- In the project root directory, create an additional directory named *dbRes*, then copy the absolute path of this directory to the clipboard.
- Open the *Database* tool window and from the top bar click the + icon. From the dropdown menu choose “Data source from Path”.
- In the dialog window, for “File name” paste the copied path to the folder and then confirm it by clicking “select folder”.
- In the next dialog window, for “Driver” choose the “H2” option.
- Set authentication to “No auth” and click Apply.
- Copy the contents of the “URL” field to the clipboard.
- Confirm creation of the database by clicking OK.
- A window named “console” should open (if it did not open automatically, in the database panel you can use the “Jump to Query Console” option or press Ctrl+Shift+F10).
- Enter the table creation statement:

```
CREATE TABLE circles (x INT, y INT, r INT, g INT, b INT);
```

and run it by clicking the play button or pressing Ctrl+Enter.
- Check the database to verify that a new table `circles` has been created, then deactivate the database connection (button “Deactivate” in the database panel or Ctrl+F2).

2. Add the class `DatabaseFeeder.java`, included in the assignment, to the project and then:

- In the `main` method, correct the parameter used when calling the method `loadFromFile`, providing the path to the file saved during labs 8 and 9 (containing the circles).
- Define a private static method `feedDB(int[] data)`, responsible for saving the data into the database in the `circles` table.

- Prepare an SQL statement for inserting records into the table `circles` with five columns: `x`, `y`, `r`, `g`, `b`. Use the statement `INSERT INTO circles (x, y, r, g, b) VALUES (?, ?, ?, ?, ?)` to allow later binding of values through `PreparedStatement`.
- Create a connection to the H2 database and a prepared statement. Use `DriverManager.getConnection("URL of your database (copied)")`.
- Create `PreparedStatement ps = conn.prepareStatement(sql)` in the same try with resources block to ensure automatic closing of resources.
- For each element of the `data` array perform decoding of the value and preparation of the data for insertion.
 - * Use the method `PositionAndColor.decode(val)` to obtain an array of coordinates and the color code.
 - * Convert the color byte value to a `Color` object using `PositionAndColor.byteToColor(res[2])`.
- Set the appropriate parameters in the `PreparedStatement`.
 - * `ps.setInt(1, res[0])` – the x coordinate.
 - * `ps.setInt(2, res[1])` – the y coordinate.
 - * `ps.setInt(3, c.getRed()), ps.setInt(4, c.getGreen()), ps.setInt(5, c.getBlue())` – the RGB values of the color.
 - * After setting the parameters, add the statement to the batch using `ps.addBatch()`.
- After finishing the loop, execute all collected statements at once.
 - * Use the method `ps.executeBatch()` to increase the efficiency of data writing.
- Implement SQL error handling in the `catch (SQLException ex)` block.
 - * In case of an error, print the stack trace using `ex.printStackTrace()`.
- Run the program and then use the console to verify whether all circles from the file have been inserted into the database. Present the obtained result to the instructor.